

Shop Assembly & Field Running Procedure - ACT-PKR-5500 Permanent Packer

Controlled Procedure ARP-PKR-5500 Rev B | Product: ACT-PKR-5500

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1. Scope

Defines shop assembly, final inspection hold points, and field running of the ACT-PKR-5500 hydraulic set permanent packer. Applies to serial numbers P55-24-0001 and later.

2. Reference Documents

- BOM-PKR-5500 Rev B
- DS-PKR-5500
- TS-100
- API 11D1
- Form F-77 / F-82

3. Equipment & Materials Required

Equipment / Material	Requirement
Torque machine, 10,000 ft-lb	Calibrated, cert < 12 mo
Hydrostatic test pump, 15,000 psi	With calibrated chart recorder
OD/ID gauges & drift bars	Per gauge list GL-55
Brass shear screws ACT-PKR-5500-09	Lot-certified, qty 10

4. Procedure

Steps marked 'QC HOLD' are mandatory hold points: work shall not proceed past a hold point without sign-off by the responsible party in the right-hand column. Record actual values on the job traveler.

4.1 Shop Assembly

Step	Action	Responsible / Sign-off
A1	Verify all components against BOM-PKR-5500 Rev B; record heat numbers of mandrel (13Cr) and load-path parts on traveler.	QC
A2	Deburr and clean all parts; verify seal bores free of defects (10x magnification).	Assembler
A3	Install element assembly; verify bonded element cure date < 24 months.	Assembler
A4	Install C-ring slips over cones; verify wicker orientation (upper slip wickers face DOWN).	QC HOLD
A5	Install 10 brass shear screws finger tight + 1/4 turn; witness mark each.	Assembler
A6	Torque body connections per TS-100 (2,800/3,000/3,200 ft-lb).	Assembler

Step	Action	Responsible / Sign-off
A7	Drift bore 2.375 in. x 42 in. drift bar - must pass free.	QC HOLD
A8	Hydrostatic body test per HTP-10: 10,000 psi, 15 min hold, < 2% drop.	QC HOLD
A9	Stencil serial number; complete Form F-77; band and crate.	QC

4.2 Field Running

Step	Action	Responsible / Sign-off
R1	Confirm casing 5-1/2 in. weight from tally is within 15.5-23 lb/ft.	Co. Man
R2	Make up packer to tailpipe/tubing; torque EUE per TS-100 (1,650-2,050 ft-lb).	Operator
R3	RIH max 2 ft/sec; record any tight spots.	Operator
R4	Space out; plug tubing below packer (pump-out plug standard).	Operator
R5	Pressure to 3,400 psi +/- 200 in 500 psi stages; hold 5 min at final stage.	Co. Man
R6	Verify set: 10,000 lbf slack-off and pick-up test, no movement.	Co. Man
R7	Increase pressure to shear pump-out plug (if fitted) per plug data sheet.	Operator
R8	Annulus test per well program; retain chart.	Co. Man

5. Notes & Cautions

- The packer is PERMANENT: verify depth twice before setting - resetting is not possible.
- Do not rotate the string while the packer is crossing liner tops (anti-preset feature is axial only).
- Report any set pressure outside 3,200-3,600 psi to ACT Engineering within 24 hr (Form F-91).

6. Sign-off Record

Role	Name	Signature	Date
Prepared	T. Okafor	/signed electronically/	2025-03-11
Checked (QA)	S. Nguyen	/signed electronically/	2025-03-11
Approved	J. Martinez, P.E.	/signed electronically/	2025-03-11